

CASEN PHOSPHO-SODA 24.4g / 10.8g oral solution

Name & Dosage Form:
CASEN PHOSPHO-SODA 24.4g/10.8g oral solution

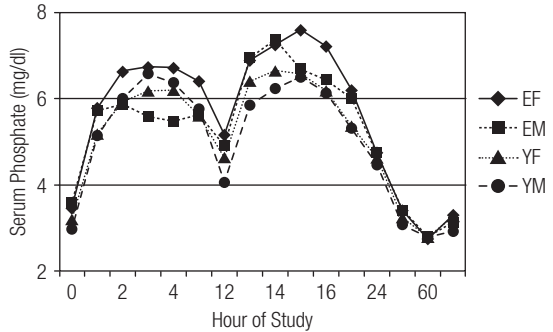
Therapeutic Class:
Laxative.
Bowel preparation.

Therapeutic Code:
A06AD17

Description:
Oral solution. Colourless solution with ginger and lime flavour without precipitation or impurities.

Names and strengths of Active Ingredients:
Disodium Phosphate dodecahydrate 10.8 g – Sodium dihydrogen phosphate dihydrate 24.4 g per 45 mL bottle.
Disodium Phosphate dodecahydrate 0.24 g – Sodium dihydrogen phosphate dihydrate 0.542 g per 1 mL.

Pharmacology (Summary of Pharmacodynamics & Pharmacokinetics):
CASEN PHOSPHO-SODA is a saline laxative that works by osmotic processes to increase fluid retention in small bowel lumen. The resultant fluid accumulation in the ileum causes distention and induces in turn bowel movement and evacuation. The start of these peristaltic movements depends on the patient and occurs 1 to 2 hours after administration.
Administration of oral sodium phosphates solution caused transient serum electrolyte shifts in healthy volunteers. An open-label study was performed with twenty-four healthy adult volunteers who received oral sodium phosphates solution to evaluate the time course and degree of electrolyte shifts in two age and two gender groups. The study was designed to mimic the bowel preparation regimen commonly used prior to colonoscopy, including a clear liquid diet, timing of sodium phosphate doses and proper hydration. The followed regimen of 2 x 45 mL of oral sodium phosphate and additional clear liquids was in line with the approved dosing regimen of the product. The study population was balanced for gender and age. One-half of the study participants were aged 65 years or older. Results showed an increase in serum concentrations of sodium and phosphate but a decrease of potassium and calcium after each dose.
The mean serum phosphate concentration for all subjects was 3.33 mg/dL at baseline, then it peaked at 6.26 mg/dL at hour 3, decreased to 4.70 mg/dL just prior to the second dose (hour 12), and peaked again at 6.86 mg/dL at hour 14. By hour 36, all serum phosphate concentrations had returned to normal. Figure shows the time-course of mean serum phosphate concentration for each age-gender subgroup. Elderly females suffered the most altered values.



Mean serum sodium concentration fluctuated within the normal range (134-147 mmol/L), however 4 subjects had sodium values above the upper limit of normal.
The fall in serum potassium and calcium concentrations fluctuated within the normal individual range and then returned to baseline values by 12 hours after administration of the second dose. 29% of subjects reported serum calcium values below the normal lower limit (8.5 mg/dL) for up to 36 hours after the administration of the first dose. Nevertheless, no clinical cases of hypocalcaemia were noted.
In conclusion, the serum electrolyte concentration shifts in healthy adults volunteers associated with the administration of 2 x 45 mL of NaP were clinically insignificant, were transient and resolved within 12 to 24 hours after completing the bowel preparation regimen.
The effect on the pharmacokinetic of Phospho-soda for patients with renal impairment has not been studied. Extrapolation of these data from healthy volunteers to at risk patients (e.g. renal patients) is not possible (See sections **Contraindications**, **Precautions/Warnings**).

Toxicology:
See Precautions/Warning.

Indications:
Laxative for cases of severe constipation. Adaptation of lifestyle and diet is also recommended.
Bowel preparation for surgery, X-rays, or endoscopy.

Recommended Dosage, Dosage Schedule & Route of Administration:
As laxative:
Posology
Adults and children over 12 years of age: 4 teaspoonfuls (20 mL).
Children aged 10 to 11 years: 2 teaspoonfuls (10 mL).
Children aged 5 to 9 years: 1 teaspoonful (5 mL).

Method of administration
It is recommended to drink one glass of water before taking Phospho-soda. Always dilute the amount indicated in a glass of cold water (±120 mL). It is recommended to drink another glass of water after the solution is administered.
Unless directed by a physician Phospho-soda should not be used for more than 3 consecutive days (see section **Precautions/Warnings**).

As a bowel cleansing product:
Posology
Two separate doses of Phospho-soda are provided to adults.

Elderly patients:
Phospho-soda should be used with caution in elderly patients. No dose adjustment is necessary in this group of patients (see section **Precautions/Warnings**).

Patients with renal impairment:
Phospho-soda is contraindicated in patients with renal impairment (see section **Contraindications**).

Patients with hepatic impairment:
The safety and efficacy of Phospho-soda in patients with hepatic impairment have not been established. Phospho-soda is contraindicated in patients with ascites (see section **Contraindications**).

Paediatric population:
Phospho-soda is contraindicated in children below 18 years (see section **Contraindications**).

Method of administration
Start to take CASEN PHOSPHO-SODA the day before hospital appointment. For appointments BEFORE 12:00 h, recommendations for MORNING appointments should be followed. For appointments AFTER 12:00 h, recommendations for AFTERNOON appointments should be followed.
The recommended dose should not be exceeded.

MORNING APPOINTMENTS:
The day BEFORE examination:
7:00 h: Drink a glass of clear liquid or water as breakfast; several glasses are recommended if possible. Clear liquid is defined as water, a diluted soup, fruit juices with no pulp, black tea or coffee, clear carbonated or non-carbonated beverages.
1st dose: Directly dilute 3 tablespoonfuls (45 mL) in half a glass of cold water (±120 mL) followed by 1 glass (240 mL) or more of cold water. Drink as much liquid as possible to replace fluid loss during bowel emptying.
13:00 h: Eat nothing for lunch. Drink instead at least 3 glasses (720 mL) of clear liquid or water. A greater volume may also be drunk.
19:00 h: Eat nothing for dinner. Drink instead at least 1 glass of clear liquid or water. A greater volume may also be drunk.
2nd dose: Directly dilute 3 tablespoonfuls (45 mL) in half a glass of cold water (±120 mL) followed by 1 glass (240 mL) or more of cold water. Drink as much liquid as possible until midnight, if needed. Drinking large amounts of clear liquid helps clean the bowel for the intervention.

AFTERNOON APPOINTMENTS:
The day BEFORE examination:
13:00 h: Lunch: eat a light snack, and then no solids until after the examination.
19:00 h: Eat nothing for dinner. Drink instead at least 1 glass of clear liquid or water. A greater volume may also be drunk.
1st dose: Directly dilute 3 tablespoonfuls (45 mL) in half a glass of cold water (±120 mL) followed by 1 glass (240 mL) or more of cold water. Drink as much liquid as possible to replace fluid loss during bowel emptying. Drink in the night at least 3 glasses (720 mL) of clear liquid or water. A greater volume may also be drunk.
On the study DAY:
7:00 h: Drink a glass of clear liquid or water as breakfast; several glasses are recommended if possible.
2nd dose: Directly dilute 3 tablespoonfuls (45 mL) in half a glass of cold water (±120 mL) followed by 1 glass (240 mL) or more of cold water. Drink as much liquid as possible to replace fluid loss during bowel emptying. Drinking large amounts of clear liquid helps clean the bowel for the intervention. Drink as much liquid as possible until 8:00 h.
This product is effective in the bowel from half an hour to 6 hours after intake.

After the procedure:
In order to replace fluid lost during the preparation for the procedure patients should be encouraged to drink plenty of fluid afterwards.

Precautions/Warnings:
CASEN PHOSPHO-SODA can only be used at the prescribed dosage and following instructions for use.

Phospho-soda has been rarely associated with severe and potentially fatal cases of electrolyte disorders in elderly patients. **The benefit/risk ratio of Phospho-soda needs to be carefully considered before initiating treatment in this at-risk population.**
Special attention should be taken when prescribing Phospho-soda to any patient with regard to known contraindications and the importance of adequate hydration. In at-risk populations (see below and sections **Recommended Dosage, Dosage Schedule & Route of Administration** and **Contraindications**) it is important to obtain baseline and post-treatment electrolyte levels.

At risk patients
Use with caution in patients with an increased risk for underlying renal impairment, pre-existing electrolyte disturbances, increased risk for electrolyte disturbances (e.g. dehydration, gastric retention, colitis, inability to take adequate oral fluid, hypertension or other conditions in which the patients are taking products that may result in dehydration; see below), hypotension with clinical impact or associated with hypovolaemia, heart disease, acute myocardial infarction, unstable angina, or with debilitated or elderly patients. In these at-risk patients baseline and post-treatment sodium, potassium, calcium, chloride, bicarbonate, phosphate, blood urea, nitrogen and creatinine values should be obtained if clinically indicated.

Dehydration
This product usually works within ½ to 6 hours. If there has been no bowel movement within 6 hours of taking Phospho-soda, instruct the patient to stop use and contact a doctor immediately as dehydration could occur.
Patients should be warned to expect frequent, liquid stools. Patients should be encouraged to drink as much liquid as possible to help prevent dehydration. Inadequate fluid intake when using any effective purgative may lead to excessive fluid loss possibly producing dehydration and hypovolemia. Dehydration and hypovolemia from purgation may be exacerbated by inadequate oral fluid intake, nausea, vomiting, loss of appetite, or use of antihypertensive drugs (e.g. angiotensin converting enzyme inhibitors (ACE-Is), angiotensin receptor blockers (ARBs), calcium channel blockers), diuretics and non-steroidal anti-inflammatory drugs (NSAIDs) and may be associated with acute renal failure. There have been rare reports of acute renal failure with purgatives, including sodium phosphates and PEG-3350. Patients with conditions that may predispose to dehydration or those taking medications which may decrease glomerular filtration rate, should be assessed for hydration status prior to use of purgative preparations and managed appropriately.

Nephrocalcinosis secondary to acute phosphate nephropathy
Nephrocalcinosis associated with acute renal failure and deposits of calcium-phosphate crystals in the renal tubules have been rarely reported in patients using sodium phosphates for bowel cleansing. Nephrocalcinosis is a serious adverse event that may result in permanent renal function impairment and the requirement of long-term dialysis. The majority of these reports occurred in elderly female patients taking drugs to treat hypertension or other drug products, such as diuretics or NSAIDs, that may result in dehydration.
Potential risk factors for acute phosphate-induced nephropathy include advanced age, inadequate hydration during use of laxatives, treatment with an ACE inhibitor, sartans, diuretics, or NSAIDs, and presence of hypertension or atherosclerosis. Care should be taken to prescribe Phospho-soda per recommendations with a particular attention to known contraindications and adequate hydration prior to, during the preparation and after the procedure and adherence to recommended spacing of doses.

Electrolyte disorders
There is a risk of elevated serum levels of sodium and phosphate and decreased levels of calcium and potassium; consequently hypernatraemia, hyperphosphataemia, hypocalcaemia, hypokalaemia, and acidosis may occur.

Hyponatraemia possibly complicated by neurological disorders, such as confusion, coma or convulsions, may occur. Slight QT interval prolongation may rarely occur as a result of electrolyte imbalances such as hypocalcaemia or hypokalaemia. These changes are clinically insignificant.

Hypomotility

Use with caution in patients with hypomotility disorders or who have had gastro-intestinal surgery or have other medical conditions predisposing them to hypomotility disorders. If the patient has had a colostomy or ileostomy, or must keep to a salt-free diet, the preparation must be used with caution, since a disturbance of electrolyte balance, dehydration or a disturbance of acid/base-balance may arise.

Lesions

Single or multiple aphthoid-like punctiform lesions located in the rectosigmoid region have been observed by endoscopy. These were either lymphoid follicles or discrete inflammatory infiltrates or epithelial congestions/changes revealed by the colonic preparation. These abnormalities are not clinically significant and disappear spontaneously without any treatment. This medicinal product contains 5 g sodium per 45 ml dose, equivalent to 250% of the WHO recommended maximum daily intake of 2 g sodium for an adult. Consideration should therefore be given to the potential harm to patients requiring a low-sodium diet. This medicinal product contains 15 mg sodium benzoate (E 211) in each 45 ml dose. This medicine contains 29 mg of alcohol (ethanol) in each 45 ml dose. The amount in 45 ml of this medicine is equivalent to less than 0.73 ml beer or 0.29 ml wine. The small amount of alcohol in this medicine will not have any noticeable effects. There are no data from the long-term use of Phospho-soda as laxative. Unless directed by a physician, Phospho-soda should not be used for more than 3 consecutive days. Frequent, long-term use of laxatives may cause habituation to laxatives and intestinal problems.

Contraindications:

Do not use:

- in children under 18 years of age for bowel cleansing;
- in the event of nausea, vomiting, and abdominal pain;
- Hypersensitivity to the active substance (s) or to any excipients listed in section **Inactive Ingredients**.

Do not use in patients with:

- renal impairment;
 - primary hyperparathyroidism associated to hypercalcaemia;
 - symptomatic heart failure (NYHA grade III or IV);
 - ascites;
 - known or suspected gastrointestinal obstruction;
 - congenital or acquired megacolon;
 - gastrointestinal perforation;
 - ileus;
 - active inflammatory bowel disease;
- CASEN PHOSPHO-SODA cannot be used in combination with other laxatives containing sodium phosphate.

Side Effects / Adverse Reactions:

The following adverse reactions were reported with frequencies corresponding to: Very common (≥1/10), Common (≥1/100 to <1/10), Uncommon (≥1/1,000 to <1/100), Rare (≥1/10,000 to <1/1,000), Very rare (<1/10,000), not known (cannot be estimated from the available data). Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness.

IMMUNE SYSTEM DISORDERS

Very rare
Hypersensitivity

METABOLISM AND NUTRITION DISORDERS

Uncommon
Dehydration

Very rare
Hyperphosphataemia
Hypocalcaemia
Hypokalaemia
Hypernatraemia
Metabolic acidosis
Tetany

Not known

Hyponatraemia complicated by neurological disorders, such as confusion, coma or convulsions

NERVOUS SYSTEM DISORDERS

Very common
Dizziness

Common
Headache

Very rare
Loss of consciousness
Paraesthesia

CARDIAC DISORDERS

Very rare
Myocardial infarction
Arrhythmia
Slight QT interval prolongation.

VASCULAR DISORDERS

Very rare
Hypotension

GASTROINTESTINAL DISORDERS

Very common
Diarrhoea
Abdominal pain
Abdominal distension
Nausea

Common
Vomiting
Colonoscopy abnormal (single or multiple aphthoid-like punctiform lesions located in the rectosigmoid region that are not clinically significant and disappear spontaneously without any treatment)

SKIN AND SUBCUTANEOUS TISSUE DISORDERS

Very rare
Dermatitis allergic

MUSCULOSKELETAL AND CONNECTIVE TISSUE DISORDERS

Very rare
Muscle cramp

RENAL AND URINARY DISORDERS

Rare
Nephrocalcinosis secondary to acute phosphate nephropaty

Very rare

Renal failure acute
Renal failure chronic

GENERAL DISORDERS AND ADMINISTRATION SITE CONDITIONS

Very common
Chills
Asthenia

Common
Chest pain

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicinal product is important. It allows continued monitoring of the benefit/risk balance of the medicinal product.

Symptoms and Treatment for Overdosage and Antidotes:

Symptoms

Fatal cases of hyperphosphataemia associated to hypocalcaemia, hypernatraemia, and acidosis has been reported when CASEN PHOSPHO-SODA is administered in excessive doses to children and patients with obstruction. In the event of overdose or accidental intake, severe adverse reactions may occur, including dehydration, hypotension, tachycardia, bradycardia, tachypnoea, cardiac arrest, shock, respiratory failure, dyspnoea, seizures, paralytic ileus, anxiety, and abdominal pain. An overdose may cause increased serum sodium and phosphate levels and decreased levels of calcium and potassium. In these cases, hypokalaemia, hypocalcaemia, hyperphosphataemia, hypernatraemia, and acidosis may occur.

Management

Cases of complete recovery after an overdose have been reported both in children accidentally given CASEN PHOSPHO-SODA and in patients with obstruction, including one with a six-fold overdose. Recovery from toxicity due to overdose may normally be achieved with rehydration, although intravenous administration of 10% calcium gluconate is required in some cases.

Interaction with other medicinal products and other forms of interaction:

Caution should be taken when patients are taking antihypertensives (e.g calcium channel blockers, angiotensin converting enzyme inhibitors (ACE-Is), angiotensin receptor blockers (ARBs)), diuretics, lithium, or other drugs that may impair electrolyte balance, because they may cause hypokalaemia, hypocalcaemia, hyperphosphataemia, hypernatraemia, dehydrationand acidosis. Use with caution in patients who are taking drugs for hyperparathyroidism. Drug absorption from the bowel may be delayed or completely interrupted when CASEN PHOSPHO-SODA is used. Activity of some commonly used oral drugs (including, amongst others, oral contraceptives, anticonvulsants, antidiabetics, and antibiotics) may be decreased or completely annulled. Administer with caution to patients taking other drugs that prolong the QT interval. Concomitant administration with other sodium phosphate preparations is contraindicated.

Fertility, pregnancy, and lactation:

Pregnancy

There are no clinical data on use of CASEN PHOSPHO-SODA in pregnant women, nor animal studies on the effect on pregnancy, embryofoetal development, parturition, and postnatal development. The potential risk for humans is not known. CASEN PHOSPHO-SODA should not be used during pregnancy unless strictly necessary.

Breast-feeding

It is unknown whether CASEN PHOSPHO-SODA is excreted in human milk. Since sodium phosphate may be excreted in human milk, it is recommended to extract and discard the milk from the first dose to 24 hours after the second dose of the bowel cleansing solution. Breast-feeding should be discontinued for 24 hours after the second dose of CASEN PHOSPHO-SODA.

Fertility

There are no data on the effect of CASEN PHOSPHO-SODA on male and female fertility.

Effects on ability to drive and use machines:

CASEN PHOSPHO-SODA may cause dizziness, possibly as the result of dehydration. This may have a minor to moderate influence on the ability to drive and use machines.

Packing / Pack sizes:

45 mL HDPE bottles.

Storage conditions and Pharmaceutical Precautions:

Store below 30°C.

Shelf-life:

3 years.

Other Information:

This medicinal product contains 5 g sodium per 45 ml dose, equivalent to 250% of the WHO recommended maximum daily intake of 2 g sodium for an adult. Consideration should therefore be given to the potential harm to patients requiring a low-sodium diet. This medicinal product contains 15 mg sodium benzoate (E 211) in each 45 ml dose. This medicine contains 29 mg of alcohol (ethanol) in each 45 ml dose. The amount in 45 ml of this medicine is equivalent to less than 0.73 ml beer or 0.29 ml wine. The small amount of alcohol in this medicine will not have any noticeable effects.

Inactive Ingredients:

Glycerol – saccharin sodium – ginger lemon flavour* - sodium benzoate – purified water.

* Ginger-lemon flavour (ginger oil, alcohol, lemon oil, citric acid, water). The ethanol content in the flavor is between 46-52% by weight.

Name and Address of Manufacturer:

CASEN RECORDATI S.L.
Autovia de Logroño, km 13,300
50180 UTEBO (Zaragoza) SPAIN

Name and Address of Importer & Distributor:

Pharmaforte (M) Sdn Bhd
2, Jalan PJU 3/49, Sunway Damansara,
47810, Petaling Jaya, Selangor Darul Ehsan, Malaysia.

Date of revision:

17 January 2022